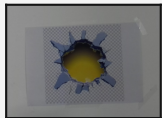
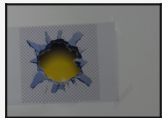


observation



left I_0



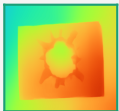
right $I_1 = \text{do}(a)$



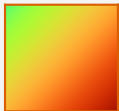
ambiguous region

hypotheses from $F_0 = E(I_0)$

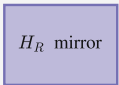
perception encoder E



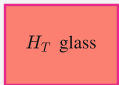
H_D direct



H_E display



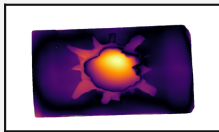
H_R mirror



H_T glass

each predicts $\hat{F}^{k,a} = W(F_t, H_k, a)$

separability under the action



$$\Delta_{DE} = \text{med } |R_D - R_E|$$

$$\mathcal{I}_A = \max_a \min_{ij} \Delta_{ij}(a)$$



ϵ

here $\mathcal{I}_A < \epsilon$

decision

$\mathcal{I}_A \geq \epsilon$: act, then answer

update $p(H_k) \propto e^{-\beta e_k} p(H_k)$
report the contact surface

$\mathcal{I}_A < \epsilon$: **ABSTAIN**

no available motion separates the hypotheses, so no answer is licensed

the model answers here regardless, and is wrong

82/296 such images are high-confidence failures